

Optimization Techniques in Power System (BT EE610)



हरियाणा केंद्रीय विश्वविद्यालय
CENTRAL UNIVERSITY OF HARYANA
NAAC ACCREDITED 'A' GRADE UNIVERSITY

Unit-III

Genetic Algorithms

By
Dr. Manish Kumar

Astt. Professor
Department of Electrical Engineering
SOE&T, CUH

What is GA?

DARWINIAN SELECTION:

From a group of individuals the best will survive

Understanding a GA means understanding the simple, iterative processes that underpin evolutionary change

GA is an algorithm which makes it easy to search a large search space

EXAMPLE: finding largest divisor of a big number

By implementing this Darwinian selection to the problem only the best solutions will remain, thus narrowing the search space.

Where GAs can be used?

OPTIMIZATION:

Where there are large solutions to the problem but we have to find the best one.

- best moves in chess
- mathematical problems
- financial problems

DISADVANTAGES

- GAs are very slow.
- They cannot always find the exact solution but they always find best solution.

Biological Background

- **Chromosome:** A set of genes. Chromosome contains the solution in form of genes.
- **Gene:** A part of chromosome. A gene contains a part of solution. It determines the solution. E.g. 16743 is a chromosome and 1, 6, 7, 4 and 3 are its genes.
- **Individual:** Same as chromosome.
- **Population:** No of individuals present with same length of chromosome.
- **Fitness:** Fitness is the value assigned to an individual. It is based on how far or close a individual is from the solution. Greater the fitness value better the solution it contains.
- **Fitness function:** Fitness function is a function which assigns fitness value to the individual. It is problem specific.
- **Selection:** Selecting individuals for creating the next generation.
- **Recombination (or crossover):** Genes from parents form in some way the whole new chromosome.
- **Mutation:** Changing a random gene in an individual.

General Algorithm of GA

START

Generate initial population.

Assign fitness function to all individuals.

DO UNTIL best solution is found

 Select individuals from current generation

 Create new offsprings with mutation and/or breeding

 Compute new fitness for all individuals

 Kill all unfit individuals to give space to new offsprings

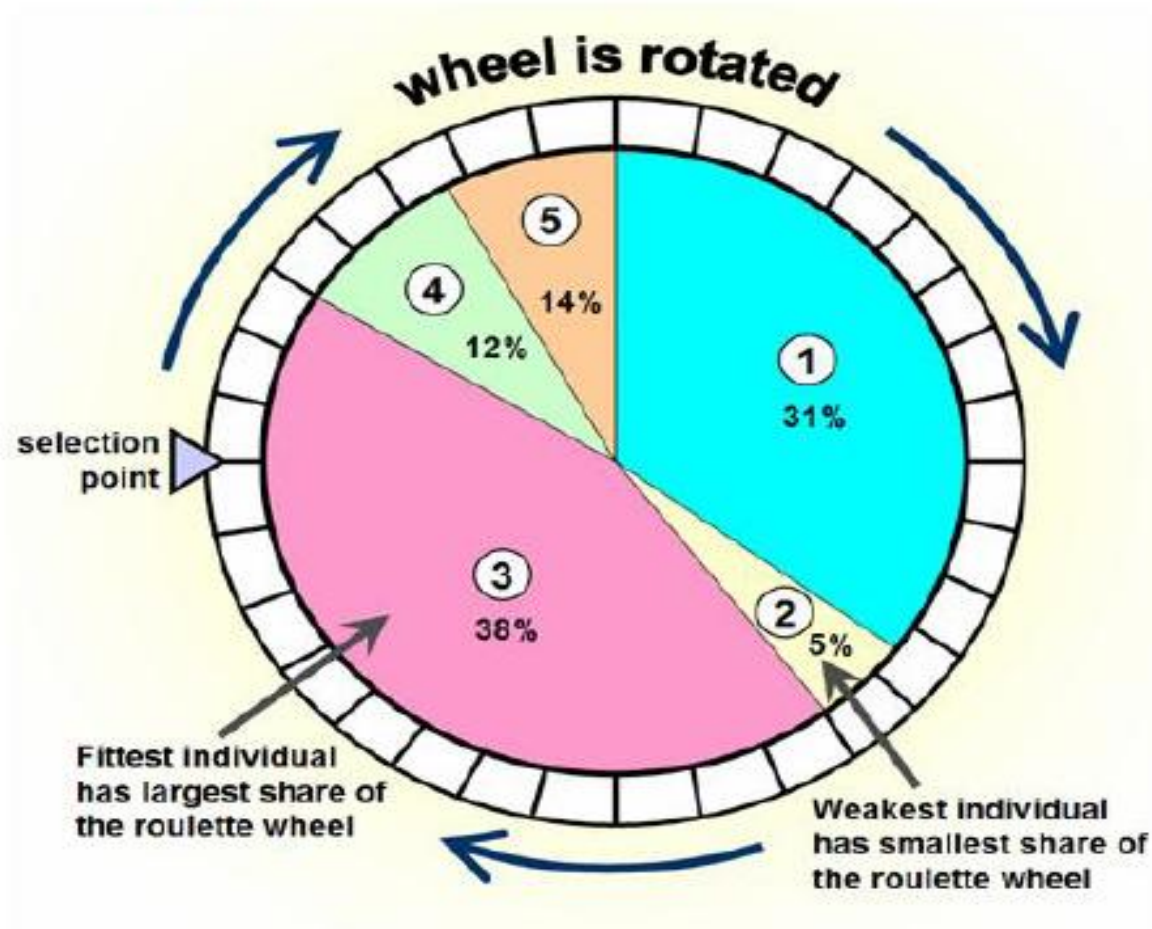
 Check if best solution is found

LOOP

END

Selection

- Darwinian Survival of The Fittest
- More preference to better guys
- Ways to do:
 - Roulette Wheel
 - Tournament
 - Truncation
- By itself, pick best



Recombination (crossover)

- Combine bits and pieces of good parents
- Speculate on new, possibly better children
- By itself, a random shuffle

Given two chromosomes

10001001110010010
01010001001000011

Choose a random bit along the length, say at position 9, and swap all the bits after that point

so the above become:

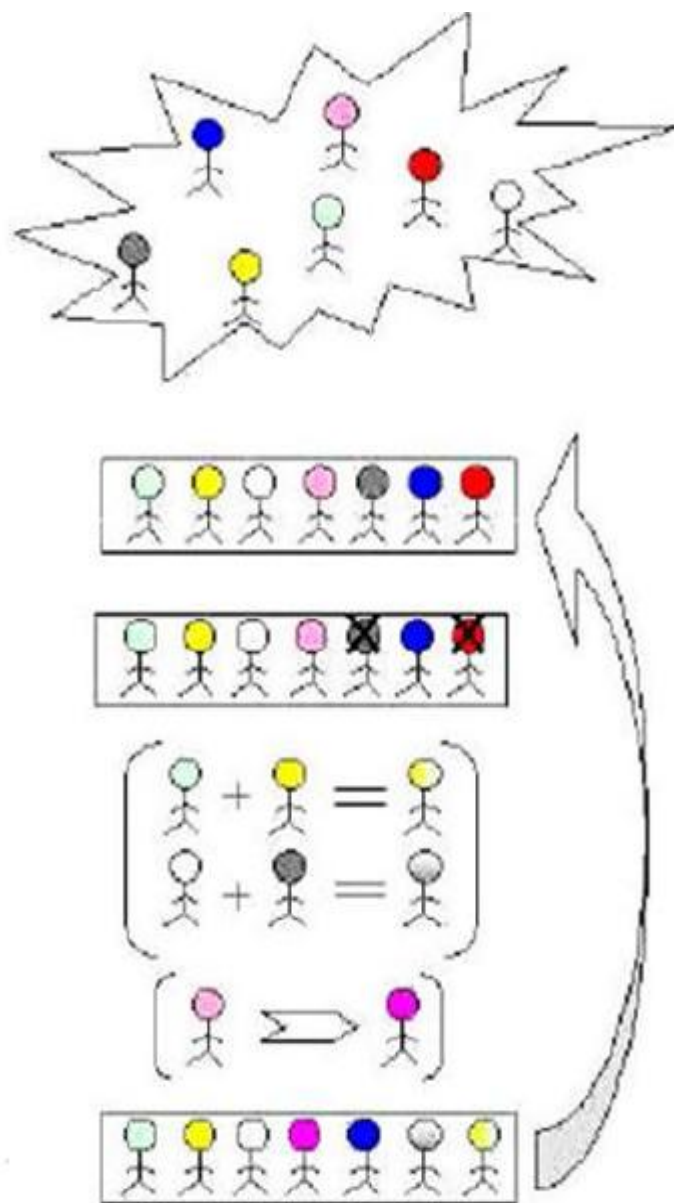
10001001101000011
01010001010010010

Mutation

- Mutation is random alteration of a string
- Change a gene, small movement in the neighbourhood
- By itself, a random walk

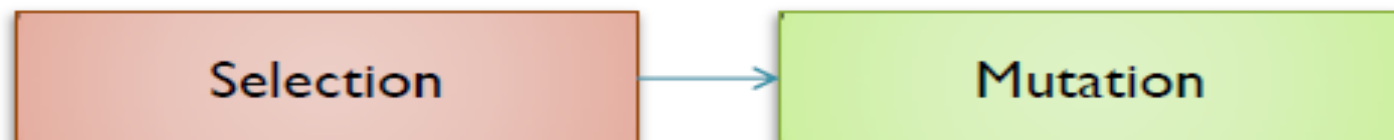
Before: **10001001110010010**

After: **10000001110110010**



Improvement / Innovation

IMPROVEMENT:



Local changes - hill climbing

INNOVATION:



Combine notions - invent

Encoding

“Coding of the population for evolution process”

BINARY ENCODING:

Chromosome A	011010110110110101
Chromosome B	101001010100101001

PERMUTATION ENCODING:

Chromosome A	1 2 3 4 5 6 7 8
Chromosome B	8 3 4 5 6 1 2 7

Example

The travelling salesman problem

Find a tour of given set of cities so that:

- each city is visited only once
- the total distance travelled is minimized

TSP – Coding for 8 cities

Encoding using permutation encoding

- | | | | |
|--------------|--------------|---------------|------------|
| 1. Chennai | 2. Trichy | 3. Thanjavur | 4. Madurai |
| 5. Bangalore | 6. Hyderabad | 7. Coimbatore | 8. Cochin |

City Route 1: (1 2 3 4 7 8 5 6)

City Route 2: (6 5 8 7 2 1 3 4)

CROSSOVER:

(1 2 3 4 7 8 5 6)
(3 1 2 4 6 5 8 7) $\longrightarrow$ (1 2 3 4 6 5 8 7)

MUTATION:

(1 2 3 4 6 5 8 7) $\longrightarrow$ (1 2 8 4 6 5 3 7)

TSP – GA Process

- First, create a group of many random tours in what is called a **population**. This algorithm uses a greedy initial population that gives preference to linking cities that are close to each other.
- Second, pick 2 of the better (shorter) tours **parents** in the population and combine them to make 2 new **child** tours. Hopefully, these children tour will be better than either parent.
- A small percentage of the time, the child tours are **mutated**. This is done to prevent all tours in the population from looking identical.
- The new child tours are inserted into the population replacing two of the longer tours. The size of the population remains the same.
- New children tours are repeatedly created until the desired goal is reached.

Survival of the Fittest

TSP – GA Process – Issues (I)

The two complex issues with using a Genetic Algorithm to solve the Traveling Salesman Problem are the **encoding of the tour** and the **crossover** algorithm that is used to combine the two parent tours to make the child tours.

In this example, the crossover point is between the 3rd and 4th item in the list. To create the children, every item in the parent's sequence after the crossover point is swapped.

Parent 1	F	A	B		E	C	G	D
Parent 2	D	E	A		C	G	B	F
Child 1	F	A	B		C	G	B	F
Child 1	D	E	A		E	C	G	D

What is the issue here ???

We get invalid sequences as children

TSP – GA Process – Issues (2)

The encoding cannot simply be the list of cities in the order they are travelled. Other encoding methods have been created that solve the crossover problem. Although these methods will not create invalid tours, they do not take into account the fact that the tour "A B C D E F G" is the same as "G F E D C B A". To solve the problem properly the crossover algorithm will have to get much more complicated.

Other Examples

THE MAXONE PROBLEM

- Suppose we want to maximize the number of ones in a string of l binary digits
- We can think of it as maximizing the number of correct answers, each encoded by 1, to l yes/no difficult questions

THE TARGET NUMBER PROBLEM

- Given the digits 0 through 9 and the operators $+$, $-$, $*$ and $/$, find a sequence that will represent a given target number. The operators will be applied sequentially from left to right as you read.

GA in Data Mining

- Used in Classification

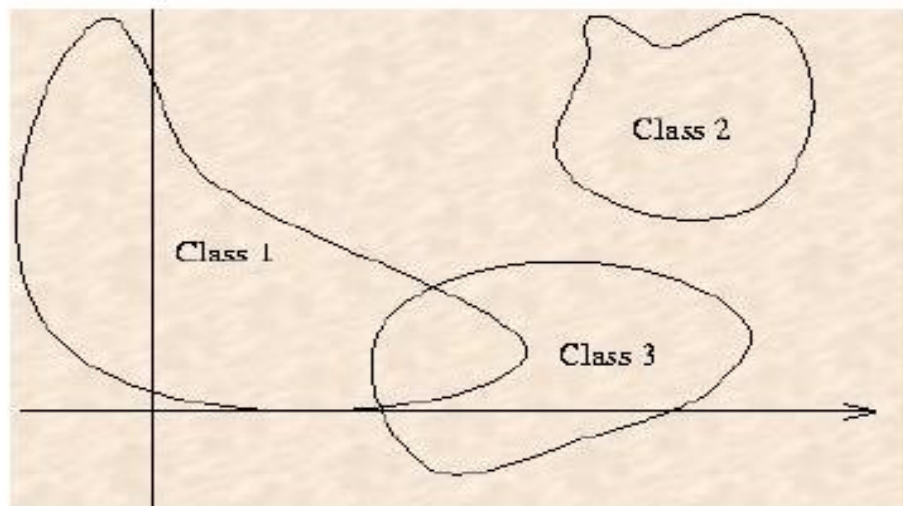
EXAMPLE:

- Two Boolean attributes, A1 and A2, and two classes, C1 and C2
- IF A1 AND NOT A2 THEN C2
100
- IF NOT A1 AND NOT A2 THEN C1
001
- If an attribute has k values, where $k > 2$, then k bits may be used to encode the attribute's values.
- Classes can be encoded in a similar fashion.

Classification Problem

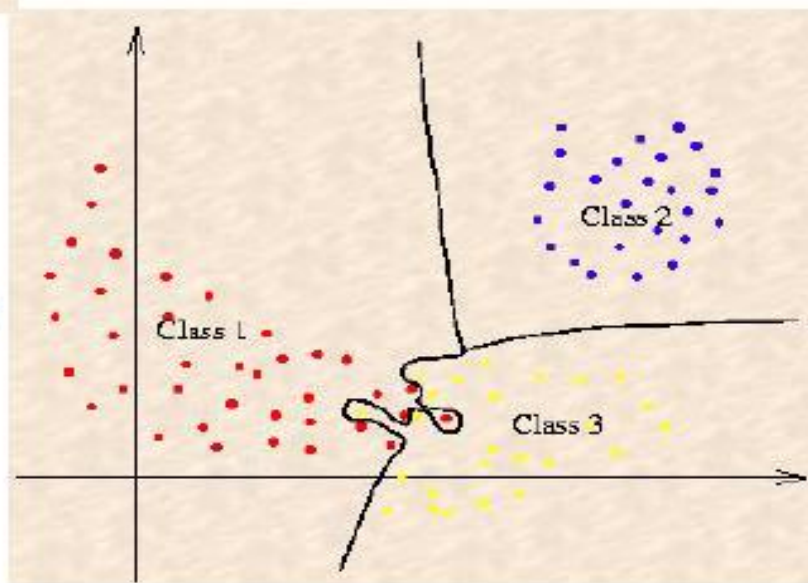
- Associating a given input pattern with one of the distinct **classes**
- **Patterns** are specified by a number of features (representing some measurements made on the objects that are being classified) so it is natural to think of them as d -dimensional vectors, where d is the number of different features
- This representation gives rise to a concept of **feature space**
- **Classification** - determining which of the regions a given pattern falls into
- A **decision rule** determines a decision boundary which partitions the feature space into regions associated with each class
- The goal is to design a decision rule which is easy to compute and yields the smallest possible probability of misclassification of input patterns from the feature space.

Classification Problem - samples



classification

An overly classified decision boundary



Discriminant Function

- **Training set** - finite sample of patterns with known class affiliations
- Use training sets to create **decision boundaries**
- Avoid over-fitting a training set by creating overly complex decision boundaries
- Simplify the shape of the decision boundary which will, by sacrificing performance on the training samples, improve the performance on new patterns
- Different classifiers can be implemented by constructing an appropriate **discriminant function $g_i(\mathbf{x})$** , where i is the class index. A pattern x is associated with the class j such that $g_j(x) > g_i(x)$ for every i not equal to j

A Linear Discriminant Function

- Linear discriminant function limits to two distinct classes
- $f(x) = \sum_{i=1}^d \omega_i x_i + \omega_{d+1}$
where x_i are the components of the feature vector and the weights ω_i need to be adjusted to optimize the performance of the classifier

HOW TO USE GA FOR CLASSIFICATION AND FINDING THE OPTIMAL WEIGHTS ω_i

- In genetic algorithms, classification problem reduces to finding the parameters of the optimum discriminant function defining the boundary between classes
- Each chromosome has a number of **genes** equal to the number of parameters used in the discriminant function
- The **fitness function** is the fraction of patterns properly classified by applying the discriminant function parameterized by the chromosome to a given testing set

Advantages of GA

- Concepts are easy to understand
- Genetic Algorithms are intrinsically parallel.
- Always an answer; answer gets better with time
- Inherently parallel; easily distributed
- Less time required for some special applications
- Chances of getting optimal solution are more

Limitations of GA

- The population considered for the evolution should be moderate or suitable one for the problem (normally 20-30 or 50-100)
- Crossover rate should be 80%-95%
- Mutation rate should be low i.e. 0.5%-1% assumed as best
- The method of selection should be appropriate
- Writing of fitness function must be accurate